Software Architecture and Design Specification

Project: Academic Performance Analytics System (APAS)
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Revision History

Version	Date	Author	Description of Changes
1.0	16-10-2026	Team 05	Initial APAS architecture and design specification.

Approvals

Role	Name	Date	Status
Prepared By	Team 05	16-10-2026	Draft
Reviewed By	To be assigned	—	Pending
Approved By	To be assigned	—	Pending

1. Introduction

1.1 Purpose

The purpose of this Software Architecture and Design Specification (SAD) is to describe the architecture, major components, design decisions, interfaces, and security mechanisms of the Academic Performance Analytics System (APAS).
This document provides a structured representation of how APAS will be designed to satisfy the functional and non-functional requirements defined in the Software Requirements Specification (SRS). It serves as a reference for developers, testers, reviewers, and other stakeholders involved in the development and validation of the system.

1.2 Scope

The Academic Performance Analytics System (APAS) is a web-based application designed to manage, analyze, and visualize students' academic performance.
The system supports three primary user roles: Administrator, Faculty Member, and Student. Its main capabilities include academic profile management, grade entry and updates, attendance tracking, performance calculations, subject- and semester-wise analysis, dashboards, graphical trends, report generation, predictive analytics, and identification of academically at-risk students.
The architecture and design must also address security, role-based access control, system performance, reliability, audit logging, and accessibility requirements specified in the APAS SRS.

1.3 Audience

This document is intended for:

- **Developers:** To understand the system architecture, components, interfaces, and design decisions.
- **Testers:** To identify components and interfaces that require verification and validation.
- **Project Team Members:** To coordinate implementation and maintain consistency across system modules.
- **Project Reviewers and Approvers:** To evaluate whether the proposed architecture and design satisfy the documented requirements.
- **Future Maintainers:** To understand the system structure and support future modifications.

1.4 Definitions

- APAS: Academic Performance Analytics System.
- Administrator: A user responsible for authorized system administration and access management.
- Faculty Member: A user who manages relevant academic records, including grades and attendance.
- Student: A user who views permitted personal academic information and performance analytics.
- RBAC: Role-Based Access Control, which restricts access according to a user's assigned role.
- SRS: Software Requirements Specification.
- SAD: Software Architecture and Design Specification.
- Audit Log: A timestamped record of relevant system activities, including academic record changes.
- Predictive Analytics: Analysis of historical academic records to estimate potential future performance.
- API: Application Programming Interface, which defines how software components communicate.

2. Document Overview

2.1 How to use this document

This document describes the architecture and design of the Academic Performance Analytics System (APAS). It is organized into sections covering the system's purpose and scope, architecture, major components, design decisions, interfaces, security, and supporting references.
Readers should use the Architecture section to understand the system structure, component responsibilities, technology choices, and security mechanisms. The Design section describes component interactions, API design, error handling, logging, monitoring, and user experience considerations.

This document should be read alongside the APAS Software Requirements Specification (SRS) to understand how the proposed architecture and design address the system's functional and non-functional requirements.

2.2 Related Documents

The following documents are related to the development, design, and testing of the Academic Performance Analytics System (APAS):

1. Software Requirements Specification (SRS) — APAS, Version 1.0, Team 05, dated 12-10-2026: Defines the functional requirements, non-functional requirements, user roles, and security requirements of the system.
2. Software Test Plan (STP) — APAS: Describes the testing scope, test strategy, test cases, test environment, and criteria for determining whether the system meets its requirements.
3. Requirements Traceability Matrix (RTM) — APAS: Maps the system requirements to their corresponding test cases to support verification and validation.

These documents serve as references for architectural decisions, component design, security implementation, and system testing.

3. Architecture

3.1 Goals & Constraints

Architecture Goals

The architecture of the Academic Performance Analytics System (APAS) aims to achieve the following goals:

1. Modularity: Separate the system into components for authentication, academic record management, attendance tracking, performance analysis, dashboards, reporting, and predictive analytics.
2. Security: Protect academic information using authentication, role-based access control, secure communication, and secure password storage.
3. Performance: Ensure that dashboards load within 3 seconds for at least 95% of requests under normal operating conditions.
4. Reliability: Achieve at least 99% monthly availability, excluding scheduled maintenance.
5. Maintainability: Design components with clearly defined responsibilities to simplify testing, debugging, and future enhancements.
6. Auditability: Maintain timestamped audit logs of academic record changes for at least 12 months.
7. Accessibility: Support keyboard navigation and text contrast in accordance with WCAG 2.1 Level AA.
8. Usability: Provide clear dashboards, graphical trends, reports, and academic insights for the different user roles.

Architecture Constraints

1. APAS must support the Administrator, Faculty Member, and Student roles.
2. Access to system functions and academic records must be restricted according to authenticated user roles.
3. Communication must use HTTPS with TLS 1.2 or higher.
4. Passwords must be stored using an approved password-hashing mechanism and must never be stored in plaintext.
5. The architecture must satisfy the functional, non-functional, and security requirements specified in the APAS Software Requirements Specification (SRS).

6. The specific technology stack and deployment environment must be confirmed by the project team

3.2 Stakeholders & Concerns

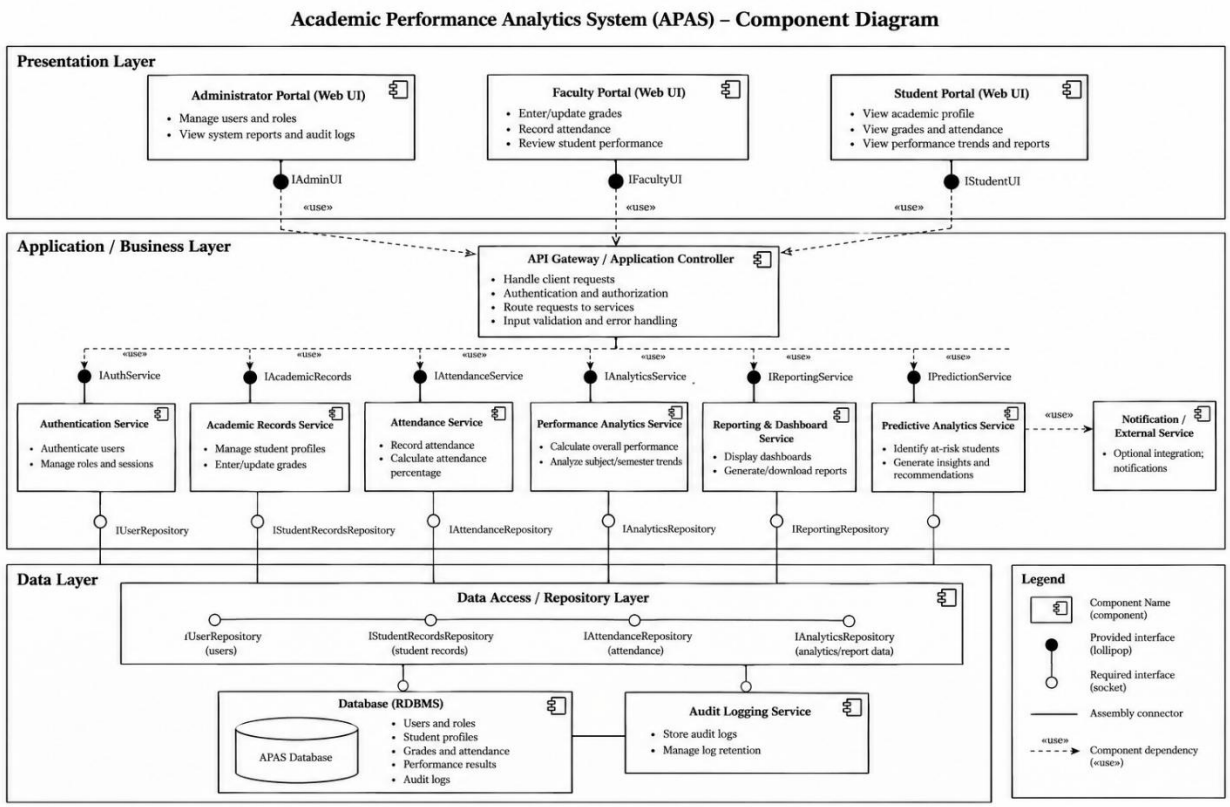
The Academic Performance Analytics System (APAS) has the following stakeholders, each with specific interests and concerns regarding the system.

1. Administrator
    - Responsibilities: Manage authorized user access and oversee system administration.
    - Concerns: Security of user accounts, correct role assignments, protection of academic records, and system availability.
2. Faculty Member
    - Responsibilities: Enter and update student grades, maintain attendance records, and review academic performance.
    - Concerns: Accuracy of academic data, ease of record management, timely performance analysis, and reliable report generation.
3. Student
    - Responsibilities: View permitted personal academic information, attendance percentages, performance trends, and relevant reports.
    - Concerns: Privacy of academic records, accuracy of displayed information, accessibility, and ease of use.
4. Project Development Team
    - Responsibilities: Design, implement, test, and maintain APAS.
    - Concerns: Clear component boundaries, maintainable code, reliable interfaces, testability, and compliance with the SRS.
5. Project Reviewers and Approvers
    - Responsibilities: Review the proposed architecture and assess whether it meets the documented requirements.
    - Concerns: Requirement traceability, completeness of the design, security controls, and satisfaction of performance and reliability targets.

Key Architectural Concerns

- Enforcing role-based access control for all protected functions and records.
- Maintaining the confidentiality and integrity of academic information.
- Providing timely dashboards and accurate academic performance calculations.
- Supporting reliable report generation and useful graphical analysis.
- Protecting audit logs and retaining academic record change history for at least 12 months.
- Supporting keyboard navigation and text contrast in accordance with WCAG 2.1 Level AA.
- Designing predictive analytics features with appropriate consideration of their limitations.

3.3 Component (UML) Diagram



3.4 Component Descriptions

The APAS architecture consists of the following major components:

1. Administrator Portal (Web UI)

Provides the interface for administrators to manage authorized user access and roles, oversee system operations, and view permitted system reports and audit logs.

2. Faculty Portal (Web UI)

Allows faculty members to enter and update student grades, record attendance, and review student academic performance.

3. Student Portal (Web UI)

Allows students to view their permitted academic profiles, grades, attendance percentages, performance trends, and reports.

4. API Gateway / Application Controller

Receives requests from the user interfaces, performs request validation, coordinates authentication and authorization checks, and routes requests to the appropriate application services.

5. Authentication Service
Authenticates users, manages user roles and sessions, and supports role-based access control to prevent unauthorized access.
6. Academic Records Service
Manages student academic profiles and supports the entry and updating of grades and other authorized academic records.
7. Attendance Service
Records student attendance and calculates attendance percentages for academic monitoring.
8. Performance Analytics Service
Calculates overall academic performance and analyzes performance by subject and semester. It supplies data for dashboards and graphical trends.
9. Reporting and Dashboard Service
Displays role-appropriate dashboards and generates reports that authorized users can view or download.
10. Predictive Analytics Service
Uses historical academic records to estimate potential academic performance, identify academically at-risk students, and provide relevant insights and recommendations. Predictions should be interpreted with consideration of model limitations.
11. Notification / External Service
Represents an optional integration point for notifications or external services, if required by the final implementation. Its use and specific interface remain to be confirmed.
12. Data Access / Repository Layer
Provides controlled access to stored user information, student academic records, attendance data, and analytics results. It separates data access responsibilities from application logic.
13. Database (RDBMS)
Stores user and role information, student profiles, grades, attendance records, and performance results. The specific database technology must be confirmed by the project team.
14. Audit Logging Service
Records timestamped events associated with academic record changes and relevant security activities. Academic record change logs must be retained for at least 12 months, as specified in the APAS requirements.

3.5 Chosen Architecture Pattern and Rationale

Chosen Architecture Pattern: Layered Architecture
The Academic Performance Analytics System (APAS) is proposed to follow a layered architecture. The system is organized into three logical layers: the Presentation Layer, the Application / Business Layer, and the Data Layer.

1. Presentation Layer
This layer provides web-based interfaces for the Administrator, Faculty Member, and Student. It displays dashboards, academic records, attendance information, performance trends, and reports according to each user's authorized access.
2. Application / Business Layer

This layer implements the main system operations, including authentication, authorization, academic record management, attendance tracking, performance analysis, report generation, and predictive analytics. It coordinates requests from the presentation layer and applies the relevant business rules.

3. Data Layer

This layer manages the storage and retrieval of user information, academic records, attendance data, performance results, and audit logs. A repository or data-access component can separate database operations from business logic.

Rationale for Choosing Layered Architecture

- **Separation of concerns:** Each layer has clearly defined responsibilities.
- **Maintainability:** Changes to one layer can be made with less impact on other layers when interfaces remain stable.
- **Testability:** Individual components and layers can be tested separately.
- **Security:** Authentication and authorization can be enforced before protected operations are performed.
- **Scalability:** Components can be optimized as usage and academic data volumes increase.
- **Traceability:** The layered design provides a clear structure for mapping system requirements to components.

A microservices architecture is not selected as the initial design because the current APAS requirements do not establish a need for independently deployed services. The layered approach provides a simpler starting point for the project.

The proposed architecture pattern should be reviewed during implementation to ensure that it continues to satisfy the requirements specified in the APAS SRS.

3.6 Technology Stack & Data Stores

Technology Stack

The Academic Performance Analytics System (APAS) is planned as a web-based application. The specific programming languages, frameworks, and development tools are yet to be finalized by the project team. The proposed technology stack should support the following requirements:

- Frontend: A web-based user interface for the Administrator, Faculty Member, and Student roles.
- Backend: Application services for authentication, academic record management, attendance tracking, performance analysis, reporting, and predictive analytics.
- Communication: HTTPS using TLS 1.2 or higher to protect data transmitted between clients and the application.
- Authentication and Security: Secure password hashing, role-based access control, and session management.
- Analytics: Facilities for analyzing historical academic records and identifying students who may be academically at risk.

Data Stores

APAS requires persistent storage for the following categories of information:

1. User and Role Data: User account details, authentication-related information, and assigned roles.

2. Student Academic Records: Student profiles, grades, subjects, and semester-related information.
3. Attendance Records: Attendance entries and calculated attendance percentages.
4. Performance and Analytics Data: Calculated academic performance, subject-wise and semester-wise analysis, and predictive analytics results where appropriate.
5. Audit Logs: Timestamped records of academic record changes and relevant security events, retained according to the requirements specified in the SRS.

A relational database management system (RDBMS) is a proposed option for storing structured academic information. The specific database product, data schema, hosting environment, and backup strategy must be confirmed by the project team before implementation.
Access to stored information must be restricted according to user roles, and appropriate controls must protect the confidentiality and integrity of academic records.

3.7 Risks & Mitigations

The following risks may affect the development, security, performance, and reliability of the Academic Performance Analytics System (APAS).

1. Unauthorized Access to Academic Records
   - Risk: Unauthorized users may gain access to confidential student academic information.
   - Mitigation: Enforce authentication, role-based access control, secure session management, and HTTPS using TLS 1.2 or higher.
2. Inaccurate or Inconsistent Academic Data
   - Risk: Incorrect grades, attendance entries, or student records may lead to misleading performance analysis.
   - Mitigation: Apply input validation, restrict record updates to authorized users, and verify performance calculations against defined rules.
3. System Performance Degradation
   - Risk: Dashboards and analytics may respond slowly as the volume of academic records or number of users increases.
   - Mitigation: Monitor response times, optimize database queries, and evaluate performance under expected workloads. The dashboard target is 3 seconds or less for at least 95% of requests under normal conditions.
4. System Downtime
   - Risk: Application or infrastructure failures may prevent users from accessing APAS.
   - Mitigation: Monitor system availability, maintain appropriate backups, and establish recovery procedures. The availability target is at least 99% per month, excluding scheduled maintenance.
5. Loss or Incomplete Retention of Audit Logs
   - Risk: Missing or incomplete logs may prevent the investigation of academic record changes and security events.
   - Mitigation: Record relevant events with timestamps, restrict access to audit logs, and verify that academic record change logs are retained for at least 12 months.

6. Misleading Predictive Analytics Results
    - Risk: Predictions based on historical academic records may be inaccurate or may incorrectly identify students as academically at risk.
    - Mitigation: Validate the predictive approach using suitable historical data, document the evaluation method and limitations, and treat predictions as decision-support information rather than guaranteed outcomes.
7. Accessibility Issues
    - Risk: Users may encounter difficulties navigating the system if the interface does not meet accessibility requirements.
    - Mitigation: Test keyboard navigation and text contrast against WCAG 2.1 Level AA requirements.
8. Unauthorized or Unintended Changes to Academic Records
    - Risk: Accidental or unauthorized modifications may compromise the integrity of academic information.
    - Mitigation: Apply role-based permissions, validate update requests, and maintain timestamped audit logs of academic record changes.

3.8 Traceability to Requirements

The following table maps the major APAS requirements to the architectural components responsible for supporting them. Requirement identifiers follow the APAS Software Requirements Specification (SRS).

Requirement ID	Requirement Description	Responsible Component(s)
APAS-F-001	User authentication and login	Authentication Service
APAS-F-002	Role-based access control	Authentication Service, API Gateway / Application Controller
APAS-F-003	Secure logout and session termination	Authentication Service
APAS-F-004	Student academic profile management	Academic Records Service
APAS-F-005	Grade entry and updates	Academic Records Service
APAS-F-006	Attendance tracking and percentage calculation	Attendance Service
APAS-F-007	Overall academic performance calculation	Performance Analytics Service
APAS-F-008	Subject-wise and semester-wise performance analysis	Performance Analytics Service
APAS-F-009	Identification of low performance and attendance below thresholds	Performance Analytics Service

APAS-F-010	Role-appropriate dashboards	Reporting and Dashboard Service
APAS-F-011	Graphical academic performance trends	Reporting and Dashboard Service, Performance Analytics Service
APAS-F-012	Report generation and download	Reporting and Dashboard Service
APAS-F-013	Predictions based on historical academic records	Predictive Analytics Service
APAS-F-014	Identification of academically at-risk students	Predictive Analytics Service
APAS-F-015	Academic insights and recommendations	Predictive Analytics Service
APAS-NF-001	Dashboard response time target	API Gateway / Application Controller, Reporting and Dashboard Service, Data Access / Repository Layer
APAS-NF-002	Monthly availability target	Application components, deployment infrastructure
APAS-NF-003	Access restrictions based on authenticated roles	Authentication Service, API Gateway / Application Controller
APAS-NF-004	Timestamped academic record change logs retained for at least 12 months	Audit Logging Service
APAS-NF-005	Accessibility requirements	Administrator Portal, Faculty Portal, Student Portal
APAS-SR-001	Valid credentials required for protected access	Authentication Service
APAS-SR-002	Role-based access control	Authentication Service, API Gateway / Application Controller
APAS-SR-003	HTTPS communication using TLS 1.2 or higher	Web interface, application hosting and communication configuration
APAS-SR-004	Approved password hashing; no plaintext password storage	Authentication Service, Data Access / Repository Layer
APAS-SR-005	Timestamped security logs for login attempts and academic record changes	Authentication Service, Audit Logging Service

This mapping provides architectural traceability between the requirements and the components responsible for implementing them. It should be reviewed and updated if the architecture or requirements change.

3.9 Security Architecture

The security architecture of the Academic Performance Analytics System (APAS) is designed to protect student academic information, prevent unauthorized access, and maintain the confidentiality and integrity of stored data.

1. Authentication

APAS requires users to provide valid credentials before accessing protected system functions. The Authentication Service verifies credentials and manages authenticated sessions. Passwords must be stored using an approved password-hashing mechanism and must never be stored in plaintext.

2. Role-Based Access Control (RBAC)

APAS supports three user roles: Administrator, Faculty Member, and Student. Access to protected functions and academic records must be restricted according to the authenticated user's assigned role.

- **Administrator:** Accesses authorized administrative functions and permitted system information.
- **Faculty Member:** Accesses authorized academic record management, grade entry, attendance management, and performance analysis functions.
- **Student:** Views permitted personal academic information, attendance, performance trends, and reports.

Authorization checks must be enforced by the application and must not rely solely on hiding options in the user interface.

3. Secure Communication

Communication between users' browsers and the APAS application must use HTTPS with TLS 1.2 or higher to protect information transmitted over the network.

4. Session Management and Logout

The system must support secure logout and session termination. After logout, the terminated session must no longer provide access to protected resources. Session handling should follow secure configuration practices.

5. Data Protection

Access to user information and academic records must be limited to authorized components and users. Input validation and appropriate access checks must be applied when academic information is created, viewed, or updated.

6. Audit Logging and Monitoring

APAS must maintain timestamped security logs for login attempts and academic record changes. Audit logs must be protected against unauthorized access or modification. Academic record change logs must be retained for at least 12 months, as required by the SRS.

7. Security Responsibilities of Components

- **Authentication Service:** Verifies credentials and manages authenticated sessions.
- **API Gateway / Application Controller:** Coordinates authorization checks and routes requests to appropriate services.
- **Academic Records Service:** Enforces permissions when student profiles and grades are accessed or modified.
- **Attendance Service:** Restricts attendance operations to authorized users.

- **Audit Logging Service:** Records relevant security events and academic record changes.
- **Data Access / Repository Layer:** Applies controlled access to stored information.

8. Security Requirements Traceability

The security architecture addresses the following APAS SRS requirements:

- **APAS-SR-001:** Valid credentials are required before protected access.
- **APAS-SR-002:** Role-based access control is enforced.
- **APAS-SR-003:** HTTPS with TLS 1.2 or higher protects network communication.
- **APAS-SR-004:** Passwords are stored using approved password hashing, never in plaintext.
- **APAS-SR-005:** Timestamped security logs record login attempts and academic record changes.

These controls must be verified during implementation and testing against the APAS SRS and the corresponding Software Test Plan (STP).

4.Design

4.1 Design Overview

The design of the Academic Performance Analytics System (APAS) follows a layered architecture to separate the user interface, application logic, and data management responsibilities. This separation improves maintainability, testability, security, and the ability to modify individual components without unnecessarily affecting other components.

The system design consists of the following layers:

1. Presentation Layer

This layer provides web-based interfaces for the Administrator, Faculty Member, and Student. It displays dashboards, academic records, attendance information, performance trends, reports, and relevant analytics according to the authenticated user's role.

2. Application and Business Logic Layer

This layer processes user requests and implements the system's core functionality. Its responsibilities include authentication, role-based access control, academic record management, attendance tracking, performance calculations, report generation, and predictive analytics. It also validates requests and ensures that users can perform only authorized operations.

3. Data Access and Data Storage Layer

This layer manages access to stored information, including user and role details, student academic records, grades, attendance data, analytical results, and audit logs. A relational database management system (RDBMS) is proposed for persistent storage; the specific database technology and schema remain to be finalized.

The design also includes supporting services for authentication, audit logging, reporting, and predictive analytics. These services work together to provide the functionality defined in the APAS Software Requirements Specification (SRS).

The design must satisfy the system's security, performance, reliability, auditability, and accessibility requirements. In particular, dashboard responses should meet the specified three-second target for 95% of requests under normal conditions, and academic record changes must be recorded in timestamped audit logs retained for at least 12 months.

Detailed interaction sequences, API interfaces, error handling, logging, monitoring, and user-interface considerations are described in the following subsections. Implementation technologies and deployment details will be finalized during further design and development.

4.2 UML Sequence Diagrams

UML sequence diagrams illustrate the interactions between APAS users and system components during important system operations. They show the sequence of requests, validations, processing steps, and responses exchanged between components.

The following sequence diagrams describe key APAS workflows:

4.2.1 User Login

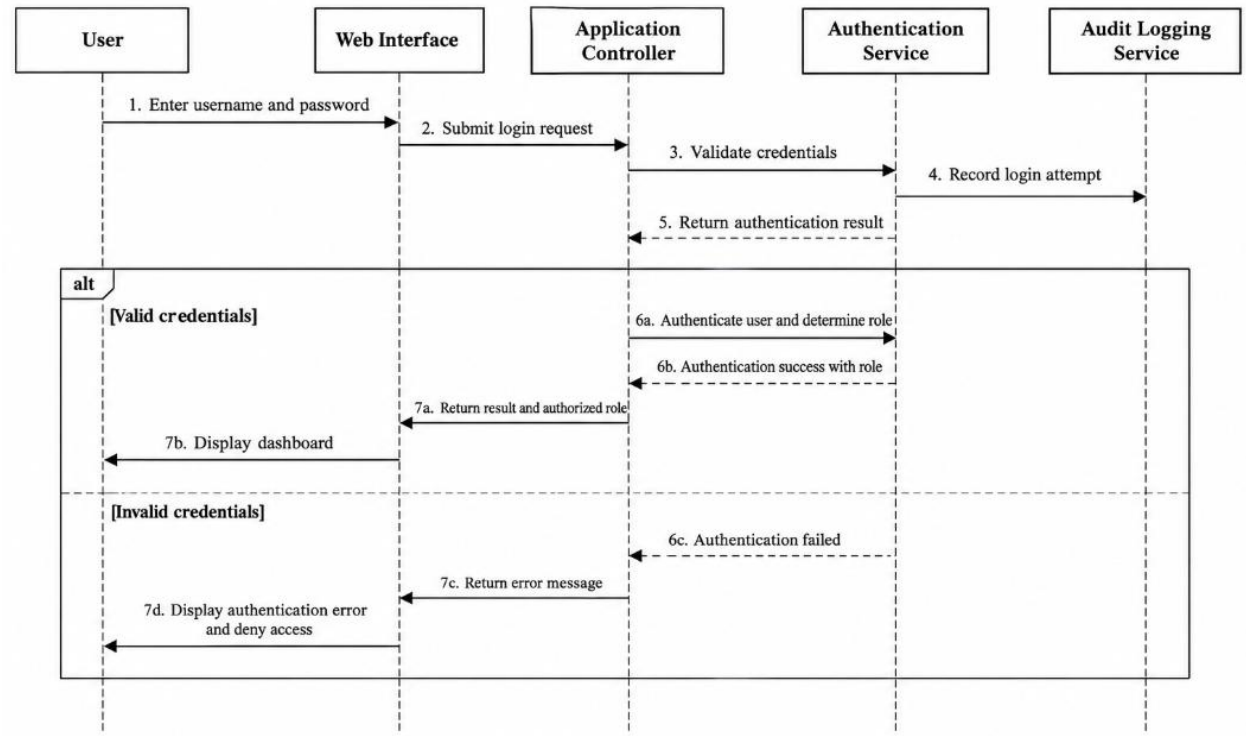


Figure 4.2.1: UML Sequence Diagram for User Login in APAS.

This sequence diagram illustrates how an Administrator, Faculty Member, or Student signs in to APAS. The user submits login credentials through the web interface. The Authentication Service validates the credentials and establishes an authenticated session when the credentials are valid. The system determines the user's role and permits access to the functions authorized for that role. If authentication fails, an appropriate error message is returned. Login attempts are recorded in the security logs

4.2.2 View Academic Performance

This sequence diagram illustrates how an authenticated user accesses academic performance information. The user requests a dashboard through the web interface. The application verifies the user's permissions and requests the relevant academic records and attendance information from the

appropriate services. The Performance Analytics Service calculates or retrieves the required performance measures, and the dashboard displays the results permitted for that user.

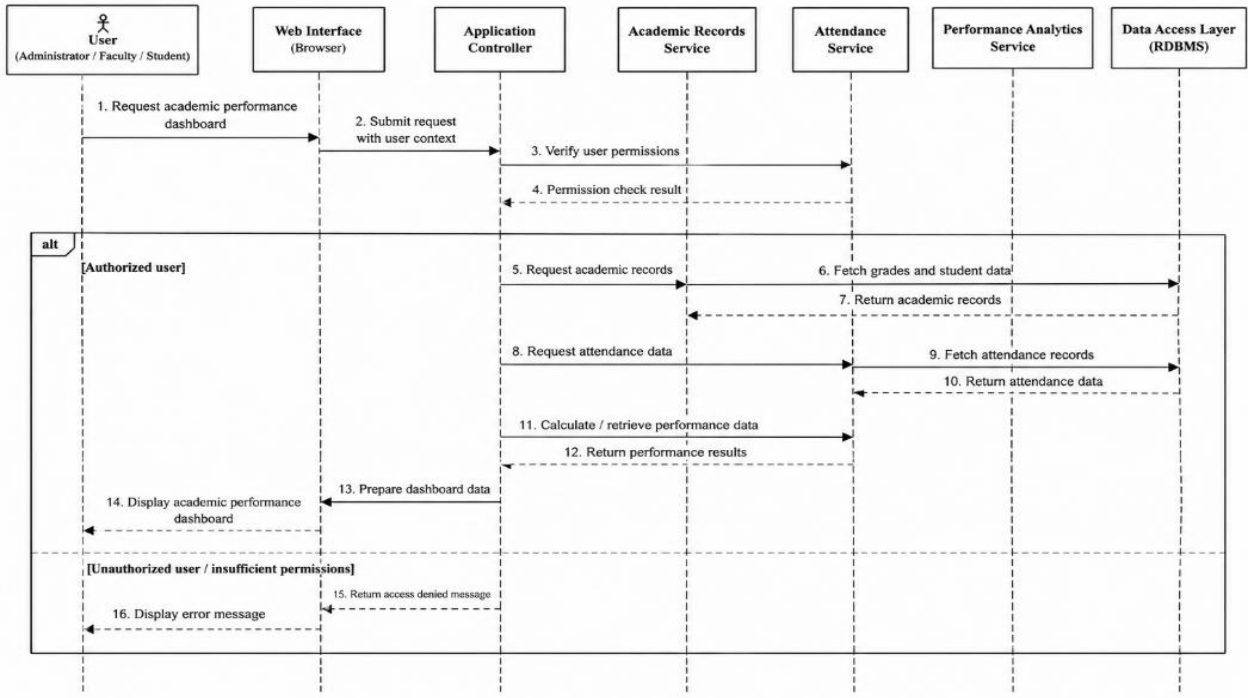


Figure 4.2.2: UML Sequence Diagram for Viewing Academic Performance in APAS.

4.2.3 Generate Performance Report

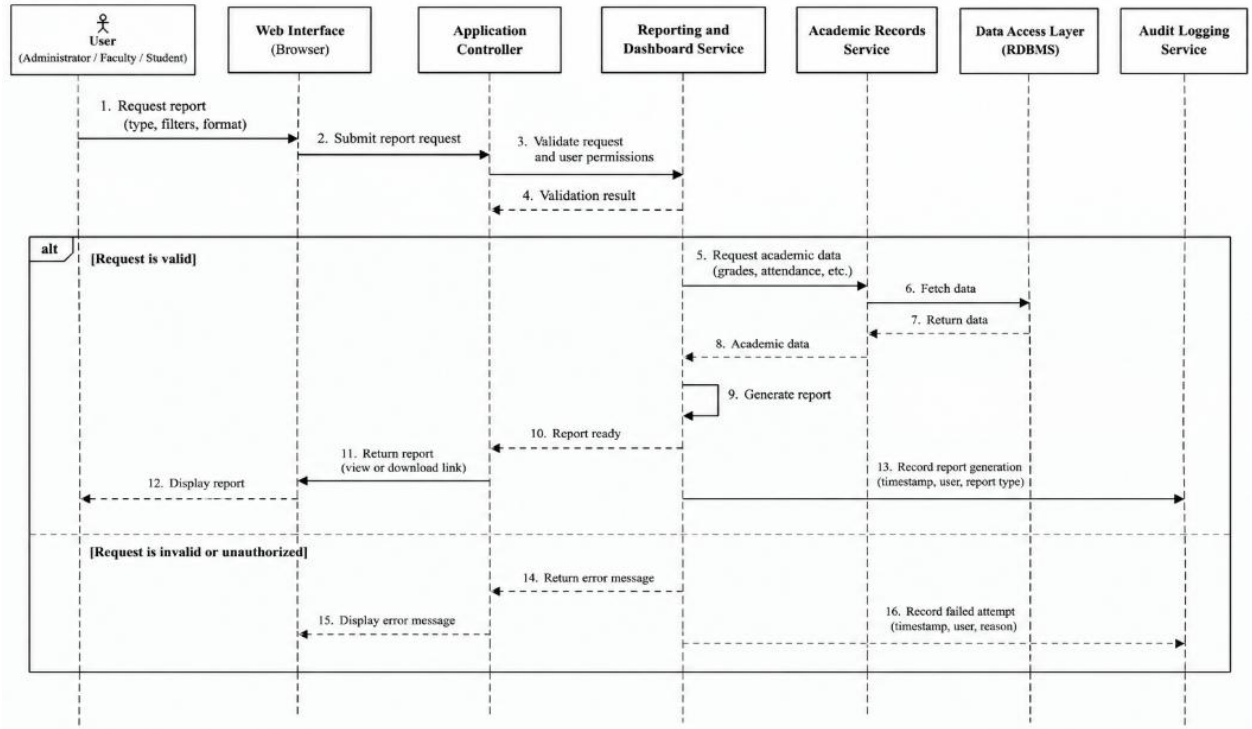


Figure 4.2.3: UML Sequence Diagram for Generating a Performance Report in APAS

This sequence diagram illustrates the report-generation workflow. An authenticated user requests a report through the web interface. The application validates the request and verifies the user's access permissions. The Reporting and Dashboard Service retrieves the authorized academic information, prepares the report, and returns it for viewing or downloading. If the request fails, the application provides an appropriate error response.
These diagrams represent the intended system design. Detailed interfaces and implementation-specific interactions will be finalized during development.

4.3 API Design

APAS will use application programming interfaces (APIs) to enable communication between the web interface and backend services. The APIs will support authentication, academic record management, attendance tracking, performance analysis, dashboard retrieval, report generation, and predictive analytics.
The API design will follow a consistent request-and-response structure. Access to protected endpoints will require authentication and role-based authorization. Requests that modify academic records will be validated, and successful changes will be recorded in timestamped audit logs.

4.3.1 Proposed API Endpoints

HTTP Method	Proposed Endpoint	Purpose	Access
POST	/api/auth/login	Authenticate a user	All users
POST	/api/auth/logout	End the authenticated session	Authenticated users
GET	/api/students/{studentId}	Retrieve a student's academic profile	Authorized users
PUT	/api/students/{studentId}	Update permitted student information	Administrator or authorized faculty
GET	/api/students/{studentId}/grades	Retrieve academic grades	Authorized users
POST	/api/students/{studentId}/grades	Enter academic grades	Authorized faculty
GET	/api/students/{studentId}/attendance	Retrieve attendance records and percentage	Authorized users
POST	/api/students/{studentId}/attendance	Record attendance	Authorized faculty
GET	/api/analytics/performance/{studentId}	Retrieve calculated performance information	Authorized users
GET	/api/dashboard	Retrieve role-appropriate dashboard information	Authenticated users

POST	/api/reports	Generate an academic performance report	Authorized users
GET	/api/analytics/predictions/{studentId}	Retrieve available performance predictions	Authorized users

These endpoints are proposed interface examples. Their final paths, HTTP methods, request formats, and permission rules will be confirmed during implementation. The endpoint list does not replace the role-based access control requirements defined in the SRS.

4.3.2 Request and Response Handling

Each API request will be validated before processing. Protected endpoints will verify the user's authentication status and permissions. Successful requests will return the requested information or confirmation of the operation. Invalid requests will return an appropriate error response without exposing sensitive information.

4.3.3 Security Considerations

All API communication will use HTTPS with TLS 1.2 or higher. Authentication credentials and session information will be handled securely. Academic record modifications will be restricted to authorized users, and relevant changes will be recorded in timestamped audit logs.

4.3.4 Error Responses

The API will use appropriate HTTP status codes to indicate request outcomes, such as:

- 200 OK — Request completed successfully.
- 201 Created — A resource was created successfully.
- 400 Bad Request — The request contains invalid input.
- 401 Unauthorized — Authentication is required or has failed.
- 403 Forbidden — The user does not have permission to perform the operation.
- 404 Not Found — The requested resource does not exist.
- 500 Internal Server Error — An unexpected server-side error occurred.

Error responses will provide useful information without exposing passwords, confidential academic data, or internal implementation details.

4.4 Error Handling, Logging & Monitoring

APAS will use a consistent approach to handling errors, recording important system events, and monitoring system performance and availability. These mechanisms will help maintain data integrity, support troubleshooting, and improve system reliability and security.

4.4.1 Error Handling

The system will validate user input and requests before processing them. When an operation fails, APAS will return an appropriate error message without exposing confidential information or internal implementation details.

The following situations will be handled:

- **Authentication errors:** Invalid credentials will result in an authentication failure message, and access to protected resources will be denied.
- **Authorization errors:** Users attempting operations outside their assigned permissions will be denied access.
- **Input validation errors:** Invalid or incomplete academic records, grades, attendance entries, and report requests will be rejected with an appropriate message.
- **Data access errors:** Database or data retrieval failures will be handled without returning incomplete or misleading results.
- **Unexpected errors:** Unexpected failures will be recorded for investigation, and users will receive a general error message that does not reveal sensitive system information.

4.4.2 Logging and Auditing

APAS will maintain timestamped logs for relevant security events and academic record changes. These logs will support security monitoring, troubleshooting, and accountability.
The logging mechanism will include:

- Login attempts, including unsuccessful attempts.
- Changes to academic records, including grades and attendance information.
- Relevant system errors and failed operations.
- Events required to investigate unauthorized access or unexpected behavior.

Academic record change audit logs will be retained for at least 12 months, as specified in the SRS. Logs will be protected against unauthorized access and modification. Passwords and other sensitive authentication information must not be written to logs.

4.4.3 Monitoring and Performance

APAS will monitor system availability, response times, errors, and important service operations. Monitoring will help identify performance degradation and service failures.
The system will be designed to meet the following requirements:

- **Dashboard performance:** Dashboards should load within three seconds for at least 95% of requests under normal conditions.
- **Availability:** The system should achieve at least 99% monthly availability, excluding scheduled maintenance.
- **Error monitoring:** Repeated failures and unexpected application errors should be recorded for investigation.
- **Auditability:** Timestamped audit records must be maintained according to the retention requirement.

Specific monitoring tools, alerting thresholds, log storage mechanisms, and operational procedures will be finalized during implementation and deployment.

4.5 UX Design

The user experience (UX) design of the Academic Performance Analytics System (APAS) will focus on simplicity, clarity, accessibility, and ease of navigation. The web interface will provide different views and functionality according to the roles of Administrator, Faculty Member, and Student.

4.5.1 Role-Based User Interfaces

Administrator Interface

The Administrator interface will support authorized administrative activities, including managing student academic profiles, overseeing academic records, and accessing relevant reports and system dashboards.

Faculty Member Interface

The Faculty Member interface will provide access to permitted academic activities, including entering and updating grades, recording attendance, reviewing student performance, and identifying students who may require academic support.

Student Interface

The Student interface will allow students to view their permitted academic profiles, grades, attendance information, performance dashboards, and relevant reports.

4.5.2 Dashboard and Data Visualization

APAS dashboards will present academic information in a clear and organized manner. Graphs and visual summaries will help users understand performance trends, subject-wise results, semester-wise progress, and attendance percentages.

Performance indicators and alerts will help authorized users identify students with low academic performance or attendance below the defined thresholds.

4.5.3 Navigation and Usability

The interface will use consistent navigation, readable labels, clear headings, and understandable feedback messages. Users should be able to access frequently used functions without unnecessary navigation.

Forms for academic records, grades, and attendance will provide input validation and meaningful error messages to reduce incorrect entries.

4.5.4 Accessibility

The interface will aim to comply with WCAG 2.1 Level AA accessibility requirements specified in the SRS. This includes keyboard navigation and sufficient text contrast. Accessibility checks will be performed during testing.

4.5.5 Responsive Design and Performance

The web interface should remain usable across supported screen sizes and devices. Dashboard interactions will be designed to meet the specified performance requirement of loading within three seconds for at least 95% of requests under normal conditions.

Detailed wireframes, final visual styles, supported screen sizes, and usability test procedures will be finalized during implementation.

4.6 Open Issues & Next Steps

Although the overall architecture and design of the Academic Performance Analytics System (APAS) have been defined, certain implementation and deployment decisions remain open. These decisions will be finalized during the detailed design and development phases.

4.6.1 Open Issues

The following issues require further clarification:

1. **Technology Stack:** The programming language, frontend framework, backend framework, and development tools must be finalized.
2. **Database Selection:** The specific relational database management system (RDBMS), database schema, and data relationships must be defined.
3. **Deployment Environment:** The hosting environment, server configuration, and deployment approach must be determined.
4. **Authentication and Session Management:** The implementation details for secure authentication, session handling, and logout must be finalized.
5. **Predictive Analytics:** The prediction methodology, input data, validation approach, and evaluation criteria must be established.
6. **Report Formats:** The supported report formats and report-generation details must be confirmed.
7. **Performance and Availability Monitoring:** The monitoring tools, alerting procedures, backup strategy, and recovery procedures must be specified.
8. **User Interface Design:** Detailed wireframes, visual styles, and usability testing procedures must be completed.

4.6.2 Next Steps

The following activities are planned for the next stages of development:

- Finalize the technology stack and development environment.
- Define the database schema and data validation rules.
- Develop the authentication and role-based access control mechanisms.
- Implement academic record management and attendance tracking.
- Develop performance analysis, dashboards, graphical trends, and report generation.
- Design and validate the predictive analytics functionality.
- Implement audit logging, error handling, and monitoring.
- Conduct functional, security, performance, reliability, and accessibility testing.
- Verify that the implemented system satisfies the requirements in the APAS Software Requirements Specification (SRS).

These activities will help ensure that APAS is implemented in accordance with its documented requirements and architectural design.

5. Appendices

5.1 Glossary

Term	Definition
APAS	Academic Performance Analytics System.
API	Application Programming Interface, which allows software components to communicate with each other.
Audit Log	A timestamped record of important system events and academic record changes.

Authentication	The process of verifying a user's identity using valid credentials.
Authorization	The process of determining which resources and operations a user is permitted to access.
Dashboard	A user interface that presents academic information and performance indicators in an organized format.
Faculty Member	A user authorized to perform permitted academic activities, such as entering grades and recording attendance.
RBAC	Role-Based Access Control, a method of assigning permissions according to user roles.
RDBMS	Relational Database Management System, used to store and manage related data.
SAD	Software Architecture and Design Specification.
SRS	Software Requirements Specification.
STP	Software Test Plan.
Predictive Analytics	The use of historical academic records to estimate potential future academic performance.
UML	Unified Modeling Language, used to represent software structure and behavior through diagrams.
WCAG	Web Content Accessibility Guidelines, which provide recommendations for accessible digital content.

5.2 Reference

[1] ISO/IEC/IEEE, *Software, systems and enterprise — Architecture description*, ISO/IEC/IEEE 42010:2022, Nov. 2022. Available: https://www.iso.org/standard/74393.html
[2] OWASP Foundation, *Application Security Verification Standard (ASVS), Version 5.0.0*, May 2025. Available: https://owasp.org/www-project-application-security-verification-standard/
[3] R. Ross, M. Winstead, and M. McEvilley, *Engineering Trustworthy Secure Systems*, NIST Special Publication 800-160, Vol. 1, Rev. 1, Nov. 2022. Available: https://doi.org/10.6028/NIST.SP.800-160v1r1

5.3 Tools

The following tools are used or intended to support the design and documentation of APAS:

1. **PlantUML:** Used to create UML diagrams, including sequence and component diagrams.
2. **draw.io (diagrams.net):** Used to create and edit architecture diagrams and other visual representations of the system.
3. **Swagger:** Used to document, describe, and support the design of API interfaces.